

Graphite-Embedded High-Performance Insulated Metal Substrate for Wide-Bandgap Power Modules

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Abstract—Emerging wide-bandgap (WBG) semiconductor devices such as silicon carbide (SiC) metal–oxide semiconductor field-effect transistors (MOSFETs) and gallium nitride high-electron-mobility transistors can handle high power in reduced semiconductor areas better than conventional Si-based devices owing to superior material properties. With increased power loss density in a WBG-based converter and reduced die size in power modules, thermal management of power devices must be optimized for high performance. This article presents a graphite-embedded insulated metal substrate (thermally-annealed-pyrolytic-graphite-embedded insulated metal substrate—IMSwTPG) designed for WBG power modules. Theoretical thermal performance analysis of graphite-embedded metal cores is presented, with design details for IMSwTPG with embedded graphite to replace a direct-bonded copper (DBC) substrate. The proposed IMSwTPG is compared with an aluminum nitride-based DBC substrate using finite-element thermal analysis for steady-state and transient thermal performance. The solutions' thermal performances are compared under different coolant temperature and thermal loading conditions, and the proposed substrate's electrical performance is validated with static and dynamic characterization. Using graphite-embedded substrates, junction-to-case thermal resistance of SiC MOSFETs can be reduced up to 17%, and device current density can be increased by 10%, regardless of the thermal management strategy used to cool the substrate. Reduced transient thermal impedance of up to 40% of dies owing to increased heat capacity is validated in transient thermal simulations and experiments. The half-bridge power module's electrical performance is evaluated for ON-state resistance, switching

performance, and switching loss at three junction temperature conditions. The proposed substrate solution has minimal impact on conduction and switching performance of SiC MOSFETs.

Index Terms—Direct-bonded copper (DBC), graphite, insulated metal substrate (IMS), integration, MOSFET, packaging, power module, silicon carbide (SiC), wide-bandgap (WBG) power devices.

I. INTRODUCTION

THE benefits of wide-bandgap (WBG) materials for power semiconductor devices are well studied in the literature. WBG materials allow power devices to operate with high efficiency at high switching frequencies, high blocking voltages, and higher junction temperatures owing to superior material properties in comparison to silicon [1]. In power electronic converters, the power devices are enclosed in discrete (e.g., TO-247) or module packages for electrical connections (e.g., power terminals), thermal management (e.g., heat sink attachment), and protection from the environment (e.g., dust and humidity). The materials used for power module packaging and the integration techniques employed for the electrical and thermal management of the emerging WBG semiconductor devices, such as silicon carbide (SiC) metal–oxide semiconductor field-effect transistors (MOSFETs) and gallium nitride (GaN) high-electron-mobility transistors (HEMTs), play a dominant role in the performance of these devices in power converters. The power module should enable heat extraction capability and low electrical parasitic inductance to maximize power handling and minimize semiconductor losses. High-performance modules are of interest in demanding application domains such as electric vehicles and more electric aircraft applications, where high power density is desired [2].

A traditional power module based on a direct-bonded copper (DBC) substrate and single-sided cooling is presented in Fig. 1. This multilayer multimaterial structure has limited heat extraction capability and targets silicon-based power modules with its low cost and maturity. Certain layers in the structure shown in Fig. 1 are subject to high mechanical stress owing to different coefficients of thermal expansion (CTEs) between layers during power and environmental cycling of the module, thus leading to limited lifetime and early failures caused by thermal stress [3]. The DBC substrate in the power module is made of a ceramic insulator sandwiched between two Cu layers. The top layer includes the tracks, die attachment, wire bond landing, and terminal attachment areas. The bottom layer includes the contact area of the substrate to the thermal management domain. The

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This article has supplementary downloadable material available at <https://ieeexplore.ieee.org>, provided by the authors. The material consists of three videos demonstrating temperature across DBC and IMSwTPG substrates during the cooldown period.

Color versions of one or more of the figures in this article are available online at <https://ieeexplore.ieee.org>.

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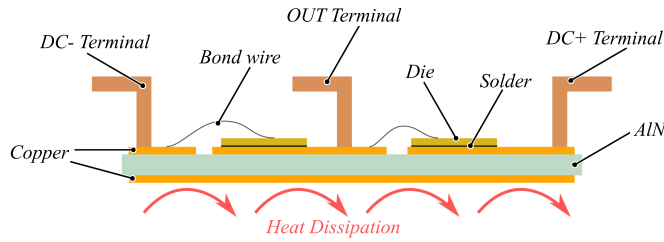


Fig. 1. Conventional DBC-based half-bridge power module [3].

TABLE I
STANDARD LAYER THICKNESS OF INSULATORS AND
FUNDAMENTAL PROPERTIES [15]

Material	Standard thickness [μm]	Heat transfer coefficient [$\text{W}/(\text{cm}^2\text{K})$]	Breakdown voltage [kV]
Al_2O_3	381	6.3	5.7
AlN	635	28.3	12.7
Si_3N_4	635	11	8.9
HT-07006	152	1.41	11

bottom layer of the DBC substrate is generally attached to a base plate for heat spreading and mounting the structure to the cold plate with a thermal interface material between the cold plate and the module. As the cooling system is grounded in almost every power electronic application, the ceramic in the DBC substrate must withstand the voltage potential between the top layer nodes and the ground. Because of its multipurpose design, the DBC substrate must provide excellent thermal and electrical performance in terms of thermal conductivity and breakdown voltage to transfer the generated heat in the dies to the heat sink and to isolate the live terminals on top of the DBC substrate from the remainder of the system.

Packaging concepts incorporate various integration architectures and advanced materials to overcome the thermal management and high electrical parasitic inductance issues to improve utilization of WBG devices. These concepts are reviewed in detail in [4] and [5]. Most of the solutions presented in these two review papers are based on DBC substrates, in which DBCs are used to provide single- and double-sided packaging solutions with the thermal and electrical functionality described in the previous paragraph. Solutions based on wire-bondless and double-sided cooling seem to promise high thermal performance and very low parasitic inductance, but these require complex module manufacturing processes and a complex cooling system. Double-sided packaging also introduces a repeatability problem, which results in possible manufacturing yield issues. Several publications [6]–[9] present direct substrate cooling for DBC-based double-sided modules. However, these solutions do not overcome the fundamental limitations of DBC, which are explained in the following paragraph.

Various ceramic insulators have been studied and are described in the literature for direct-bonded substrates. Fundamental properties of the three most common ceramic materials used in industry are presented in Table I. Aluminum oxide (Al_2O_3) is the commonly used insulator in substrate design for cases

in which the mechanical strength and relatively a high CTE are desired and power density is not a concern. The manufacturing of Al_2O_3 is relatively cheaper than aluminum nitride (AlN) owing to the existence of an oxide layer on the ceramic that forms the bonding between the metal and the ceramic. Although the standard thickness for Al_2O_3 is presented as 381 μm in Table I, 500 and 630 μm values are also encountered for high-voltage designs. AlN is the high-performance dielectric option among all the ceramics presented in Table I because of its high thermal conductivity and breakdown voltage. AlN is two to three times more expensive than Al_2O_3 , and it is also known to be a brittle material. The standard thickness of AlN is stated as 630 μm , and 1-mm thickness is used in very high voltage applications. Because of the mechanical strength and brittle structure of AlN, a 630- μm -thick insulator is commonly used for applications in which 2–3-kV isolation is required. In contrast, silicon nitride (Si_3N_4) is mechanically stronger than AlN, so it is possible to machine Si_3N_4 thinner than the standard thickness of 630 μm . This presents the opportunity to match the thermal performance of AlN-based DBC at low-voltage applications using Si_3N_4 without compromising the electric isolation performance. Reliable high-performance Si_3N_4 -based DBCs have been reported in literature [10].

The alternative to using DBC for power electronic applications is to use an insulated metal substrate (IMS), in which a polymer-based film is used to insulate the conductive layers in the substrate. IMSs have been studied for power modules [11]–[14], and the thermal performance of IMS presented in these works is usually lower than that of the conventional DBC solutions. The new dielectric polymer/ceramic blend HT-07006, which was introduced by Henkel and is presented in Table I, has excellent dielectric strength and can be processed in very thin layers. These properties make it an attractive solution for high-power applications, despite the lower thermal performance with respect to ceramic materials. In addition to this, polymer materials have a wide elastic deformation range, and the thermal expansion performance is not of interest in substrate applications [15]. In recent papers [16], [17], Gurpinar *et al.* demonstrate that the thermal performance of the IMS can match that of an AlN-based DBC substrate by providing a copper layer with adequate thickness underneath the power semiconductor dies. The thick copper layer provides lateral heat spreading and increase in the effective heat transfer area for SiC MOSFETs. The IMS also allows for the possibility of making multilayer substrates for an optimized electrical layout, unlike in DBC substrates. Previous work on IMS development for power modules shows that the structure requires further improvement to make it viable and attractive for emerging SiC- and GaN-based power devices.

Graphite has gained attention for thermal management of electronics in recent years because of high thermal conductivity and low density in comparison to commonly used metals such as copper and aluminum. Advanced thermal interface materials have been created and evaluated for the purpose of better heat transfer; these new interface materials are infused with graphite and especially made to assist in heat spreading, producing up to a 50% increase in heat dissipation between layers of a power package [18]. Another method involving the use of graphite

TABLE II
TYPICAL PROPERTY COMPARISON OF HEAT SPREADER MATERIALS

Material	In-plane TC $\left[\frac{W}{m \cdot ^\circ K}\right]$	Cross-plane TC $\left[\frac{W}{m \cdot ^\circ K}\right]$	In-plane CTE $\left[\frac{ppm}{^\circ C}\right]$	Specific gravity	In-plane TC Specific gravity
Aluminum	218	218	23	2.7	81
Copper	396	396	17	8.9	45
AlSiC-12	180	180	11	2.9	62
CuW	185	185	8.3	15.2	12
TPG	> 1500	10	-1	2.3	650

deals with especially made heat sinks; these heat sinks yielded a 16% decrease in the maximum temperature when compared with aluminum [19]. In [20], a highly oriented pyrolytic graphite heat spreader is investigated for microelectronic assemblies. The graphite was embedded in an aluminum heat spreader and performed 50% better than the aluminum-based heat spreader. Natural-graphite-sheet-based heat sinks for air-cooled power electronic converters are investigated by Cermak *et al.* [21]. The graphite-sheet-based heat sink reduced thermal resistance by 31% and diode junction temperature by 19%. Furthermore, the presented graphite-based heat sink was 55% lighter than the aluminum-based heat sink. The impact of composite heat spreaders formed by highly oriented pyrolytic graphite sandwiched between two copper layers is investigated in [22], in which Bostanci and Obuladinne showed that a composite heat spreader using highly oriented pyrolytic graphite provides uniform heat flux and temperature profile at the cooling surface, therefore enabling different cooling options for thermal management of devices with extreme heat fluxes, such as GaN HEMTs and X-ray tubes. Finally, in [23], the impact of composite heat spreaders and heat sinks formed by encapsulated of thermally annealed pyrolytic graphite (TPG) in tin for heat sink and in aluminum for heat spreader were discussed. The authors showed a twofold increase in LED power in comparison with the aluminum-based baseline design.

This article proposes a graphite-embedded multilayer IMS that can outperform conventional DBC substrates. The rest of this article is organized as follows. Section II presents the fundamental thermal properties of graphite-embedded copper. In Section III, design and construction details of the proposed IMS are presented, along with the details of the DBC substrate that is used to compare the performance of the proposed solution. Steady-state and transient thermal simulation results of the proposed structure and comparison with DBC based on finite-element analysis (FEA) are presented in Section IV. The experimental setup for thermal and electrical analysis, experimental results including steady-state and transient thermal analysis, and static and dynamic electrical characterization of the SiC-MOSFET-based WBG power module are discussed in Section V. Finally, Section VI concludes this article.

II. THERMALLY ANNEALED PYROLYTIC GRAPHITE

TPG is an advanced thermal management material formed by well-aligned graphene planes that offer very high in-plane

thermal conductivity. Graphene layers are processed through a high-temperature chemical vapor deposition process and heat treatment above 3000 °C to form a well-aligned oriented multilayer graphite structure [24]. TPG is a relatively soft material owing to the weak Van Der Waals force between the graphene layers. The weak bond between those layers results in low cross-plane thermal conductivity. Furthermore, the inertness of graphite prevents it from direct joining to other metals or ceramics during regular plating, soldering, and brazing processes. Therefore, the graphite is encapsulated in a metal to protect the relatively brittle material and to make it available for attachment to other materials.

The comparison of the typical properties of heat spreader materials presented in Table II shows that TPG provides up to four times higher in-plane thermal conductivity (>1500 W/m·K) with one-fourth of the weight compared to copper, which is commonly used in thermal management of power modules such as DBCs and base plates. However, encapsulating TPG in a metal case results in higher thermal conductivity than the metal encapsulant, but lower than pure TPG material. Hence, the ratio of graphite to metal directly affects the thermal performance of the structure [25]. The typical encapsulation of TPG with copper encapsulant—in which graphene layers are oriented for optimum one direction in-plane and through-plane thermal conductivity—is presented in Fig. 2. If contact resistance between the layers shown in this figure is omitted, then the in-plane and through-plane thermal conductivity of the encapsulated TPG can be calculated using the following equations [26]:

$$k_{\text{in-plane}} = \frac{t_{\text{TPG}} \cdot k_{\text{TPG}} + t_{\text{metal}} \cdot k_{\text{metal}}}{t_{\text{total}}} \quad (1)$$

$$k_{\text{through-plane}} = \frac{t_{\text{total}}}{\frac{t_{\text{TPG}}}{k_{\text{TPG}}} + \frac{t_{\text{metal}}}{k_{\text{metal}}}} \quad (2)$$

where t is the thickness of the material in a given plane and k is the thermal conductivity. Similarly, the total specific gravity (SG) of the structure, which is the ratio of the density of the structure to the density of water, can be calculated using the following equation:

$$\text{SG}_{\text{total}} = \frac{t_{\text{metal}} \cdot \text{SG}_{\text{metal}} + t_{\text{TPG}} \cdot \text{SG}_{\text{TPG}}}{t_{\text{total}}} \quad (3)$$

The thermal conductivity of the TPG-embedded structure can be calculated by applying the material properties given in Table II to (1) and (2). The in-plane and through-plane thermal conductivity

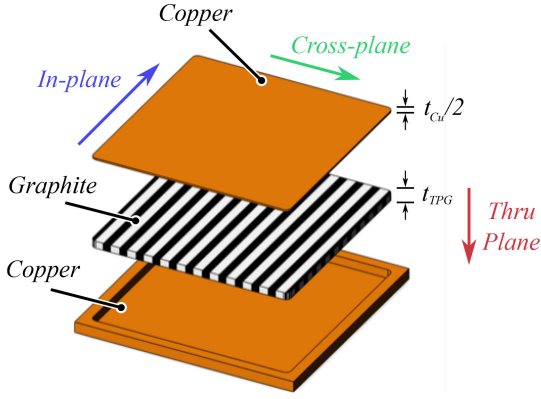


Fig. 2. TPG-embedded structure with copper encapsulant for in-plane and through-plane thermal conductivity and SG analysis.

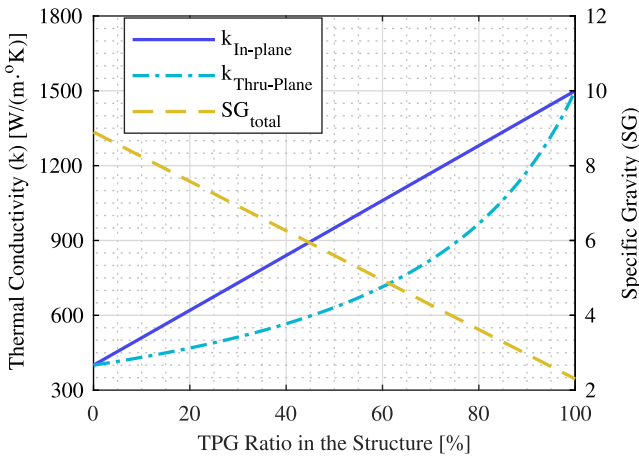


Fig. 3. In-plane and through-plane thermal conductivity and the SG of the structure in Fig. 2 with respect to the TPG ratio.

values of the structure with respect to the TPG ratio are presented in Fig. 3. In this scenario, the TPG orientation provides high thermal conductivity in both in-plane and through-plane axes. It can be seen that above 80% TPG ratio, the in-plane k is 3.2 times higher and the through-plane k is 2.4 times higher than the k of Cu. Moreover, the SG of the structure can be reduced by 2.46 times, leading to a lighter structure with improved thermal conductivity. In such a system, although the thermal conductivity and the SG of the structure are determined by the TPG ratio, as shown in Fig. 3, the CTE of the structure is determined by the CTE of the encapsulant material. In [25], TPG-embedded copper samples were tested in a thermal cycling chamber to evaluate the impact of thermomechanical stress to the thermal conductivity. The samples were cycled from -65 to 150 °C for 1000 cycles, and there was no change in thermal conductivity.

According to this theoretical analysis, with a high TPG ratio in a composite structure formed by copper and TPG, very high thermal conductivity in the through-plane and heat spreading in the in-plane can be achieved. This can lead to high-performance IMSs, where copper and aluminum are used as heat spreaders in conventional designs. The anisotropic thermal performance of the composite structure presented in Fig. 2 can be utilized in multidie substrate design for power modules, where the through-plane can provide maximum heat transfer from the each

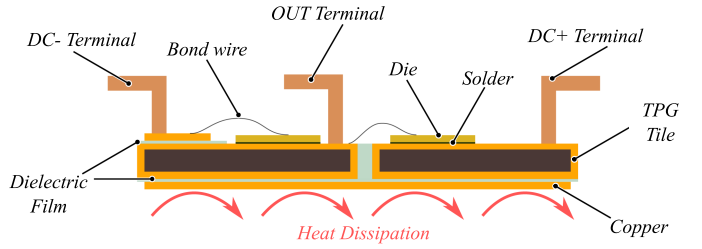


Fig. 4. Proposed IMSwTPG-based half-bridge power module.

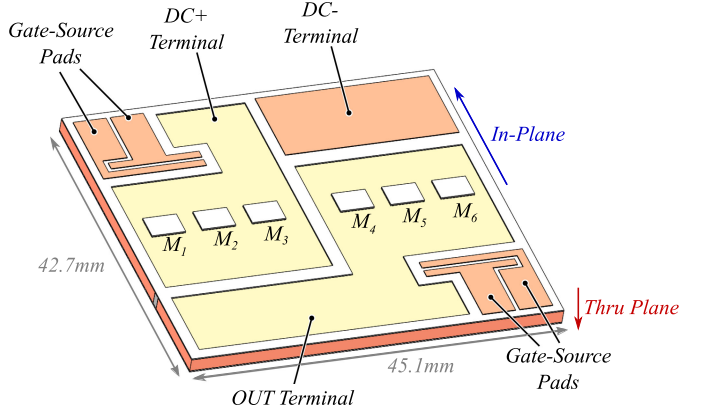


Fig. 5. Layout of the substrate for SiC-MOSFET-based half-bridge circuit.

individual die to the coolant, the cross-plane can limit the thermal cross-coupling between the paralleled dies, and the in-plane can provide lateral heat spreading to the areas in the module where heat loading is minimal.

III. SUBSTRATE DESIGN AND CONSTRUCTION

In high-power three-phase power modules, it is common to design a substrate to accommodate a single half-bridge circuit and then mount the substrates for each phase on a common heat spreader [27]. The conventional DBC substrate used for the half-bridge designs is taken as the baseline. The fundamental assumptions used for the design are the following: 1) multiple dies in parallel for high current rating per switch; 2) single-sided cooling; and 3) same breakdown voltage rating for the proposed design and the conventional DBC substrate. The DBC substrate used for the comparison in this study is formed by a $640\text{-}\mu\text{m}$ -thick AlN ceramic insulator sandwiched between copper planes that are $300\text{ }\mu\text{m}$ thick. The semiconductor dies are soldered on different top copper islands and are then connected via aluminum wire bonds to form a half-bridge power module. As mentioned in Section II, AlN is the high-performance ceramic commonly used for high-power DBC substrates, and it can be manufactured at limited thickness and size. The thicknesses of copper and the ceramic selected in this design represent the common thicknesses used in typical DBC-based designs.

For the TPG-embedded insulated metal substrate (IMSwTPG), a three-layer structure is proposed as shown in Fig. 4 to match the layout on the top layer of the DBC substrate, as presented in Fig. 1. The layout of the proposed IMSwTPG can be seen in Fig. 5, and the expanded structure is presented in Fig. 6.

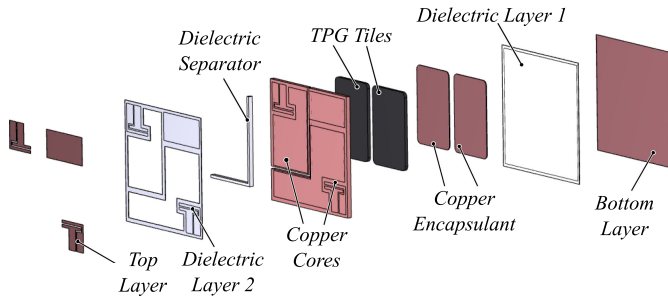


Fig. 6. Expanded structure of the IMSwTPG.

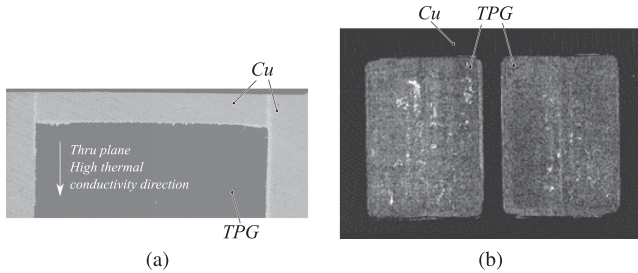


Fig. 7. TPG-embedded copper core. (a) Interface cross section (scanning electron microscopy). (b) Interface top view (ultrasound imaging c-scan).

The fundamental objective of the IMSwTPG is to spread the heat generated in the SiC MOSFET dies across the TPG-embedded Cu core to utilize the surface area of the Cu core before reaching the dielectric layer in the thermal management network. This will enhance the heat transfer area for the substrate and hence reduce the junction-to-case thermal resistance of SiC MOSFETs. For this purpose, the anisotropic properties of TPG discussed in Section II are utilized to spread the heat by aligning high-thermal-conductivity planes of the TPG tiles with the in-plane and the through-plane shown in Fig. 5. The in-plane alignment provides spreading of the heat generated in SiC MOSFET dies to the power and gate terminal areas on the substrate, where the heat flux is relatively low. From the design perspective, this is achieved by expanding the TPG tile underneath the terminal regions using the three-layer structure in the IMSwTPG, as shown in Figs. 4 and 6. Furthermore, the proposed alignment of the TPG tile will limit the heating interaction between paralleled dies owing to low thermal conductivity in that axis.

For the construction of the IMSwTPG, first, the Cu core with 1.10-mm TPG inserts and 0.25-mm-thick Cu skins are constructed to ensure excellent thermal and mechanical joints between the TPG insert and the Cu enclosure [25]. This provides approximately 68% TPG ratio in the through-plane and a thermal conductivity of 760 W/m·K (measured using Netzsch Nanoflash 467) through the Cu/TPG/Cu structure, which is almost twice that of copper's value. The interface cross section and the interface top view of the Cu/TP/Cu structure, presented in Fig. 7(a) and (b), show the hermetic and nearly void-free encapsulation. These two copper cores are isolated from the bottom copper layer using the HT-07006 dielectric film, and they are isolated from each other with nonconductive epoxy-based filler. The total

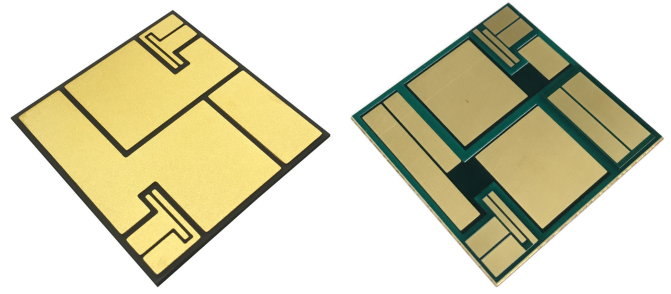


Fig. 8. Manufactured (left) AlN-based DBC and (right) IMSwTPG samples.

thickness of the copper core is 1.6 mm, and the dielectric film's thickness is 152 μm , which has the same breakdown voltage of the 640- μm AlN used in DBC. The thickness of the bottom copper layer is 70 μm . Once the copper cores, dielectric layer 1, and bottom layer are bonded, additional layers can be added to the structure. In this case, dielectric layer 2 and the top layer are added to the design to accommodate isolated electrical terminals for gate-source Kelvin connections of the SiC MOSFETs and a dc terminal for the half-bridge design. Manufactured DBC and IMSwTPG samples are presented in Fig. 8. As shown in Figs. 5 and 8, three SiC MOSFET dies are paralleled per switch in the half-bridge configuration. The SiC MOSFET dies used in this work were manufactured by Wolfspeed and were rated at 900 V, 30 m Ω (CPM3-0900-0030A [28]) at room temperature, with a dimension of 4.08 mm $\times$ 3.10 mm.

IV. FINITE-ELEMENT ANALYSIS

The FEAs of thermal performance for the DBC and IMSwTPG substrates are conducted based on representative electrical and thermal operating conditions of a liquid-cooled power module in transportation applications. The base temperature of the substrate in the thermal model is set to 65 $^{\circ}\text{C}$, which is the typical coolant temperature in automotive systems [29]. The heat transfer coefficient between the coolant and the base of the substrate was defined as 5000 W/(m 2 ·K) to represent the cooling performance of a typical liquid-cooled cold plate and the interface thermal resistance between the substrate and the cold plate [6]. The die attach is assumed to be 50 μm thick and 63-Sn/37-Pb-based solder.

For the meshing of the substrates, instead of a conventional tetrahedral mesh, a free triangular mesh was applied at the top surface of the system and distributed to multiple layers in the heat transfer axis (through-plane). This ensured that there were enough nodes in the through-plane without constraining the maximum feature size to a very low number. On the die and substrate surface, maximum feature sizes of 0.2 and 0.5 mm were used. The surface mesh was then distributed to five layers in the die, five layers in the die attach, 20 layers in the copper core, five layers in the dielectric layer, and five layers in the bottom layer to capture heat spreading and temperature distribution across the substrate under test with high accuracy and resolution. Similar settings were applied to the DBC model for consistency.

A. Steady-State Analysis

The steady-state analysis was conducted with constant power loss across individual SiC MOSFET dies and under a constant cooling profile at the base of the substrate. The constant power loss across each die was set to 45 W to represent the operating condition of the power module as a half-bridge inverter with a 100-A_{rms} load current, 600-V dc-link voltage, and 30-kHz switching frequency. These conditions were based on the operation of the 100-kW three-phase segmented inverter topology for traction drive systems, as discussed by Gurpinar and Ozpineci [30].

The steady-state thermal analysis results for DBC and IM-SwTPG, including surface temperature distribution and isothermal contours within the substrate, are presented in Fig. 9. DBC provides symmetrical temperature distribution between paralleled dies M_{1-3} and M_{4-6} , with a 144 °C maximum temperature on the middle dies M_2 and M_5 . The surface temperature distribution shown in Fig. 9(a) indicates that the heat spreading across the substrate is relatively limited. The temperature distribution within the DBC substrate presented in Fig. 9(b) shows that the high-temperature region is constrained around the SiC MOSFET dies owing to the high thermal conductivity of AlN and the limited thickness of top- and bottom-layer Cu. In contrast, the steady-state surface temperature distribution of the IMSwTPG shows that a reduced junction temperature can be achieved for dies M_{1-3} and M_{4-6} under the same operating conditions. The isothermal contours for temperature distribution within the IMSwTPG in Fig. 9(d) show the strong effect of the high thermal conductivity of TPG for in-plane heat spreading and through-plane heat transfer. TPG provides maximum utilization of the surface area underneath the SiC MOSFET dies and increases the effective heat transfer area before reaching the dielectric layer in the thermal path. Although the TPG tile size and placement within the copper core is symmetrical for M_{1-3} and M_{4-6} , the additional copper area used for the output terminal, as shown in Figs. 5 and 6, provides enhanced heat spreading for dies M_{4-6} . It should also be noted that the dielectric separator used between the copper cores shown in Fig. 6 is acting as a thermal insulator owing to the low thermal conductivity of the material. Therefore, there is approximately a difference of 12 °C in the maximum junction temperature between M_{1-3} and M_{4-6} .

B. Transient Analysis

The substrate design for the SiC MOSFET dies can have a strong impact on the junction-to-coolant thermal impedance. To evaluate the impact of the proposed IMSwTPG structure, time-dependent thermal simulations were conducted for transient thermal impedance extraction of each die on DBC and IMSwTPG. The steady-state results presented in Fig. 9 were taken as the initial condition for the time-dependent simulation. The power loss across each die was set to zero, and the base temperature was set to 65 °C, with 5000 W/(m²·K) cooling performance. The transient thermal impedance Z_{th} for each die was calculated as follows:

$$Z_{th} = \frac{\Delta T_j}{P_{Loss}} = \frac{T_j - T_c}{P_{Loss}} \quad (4)$$

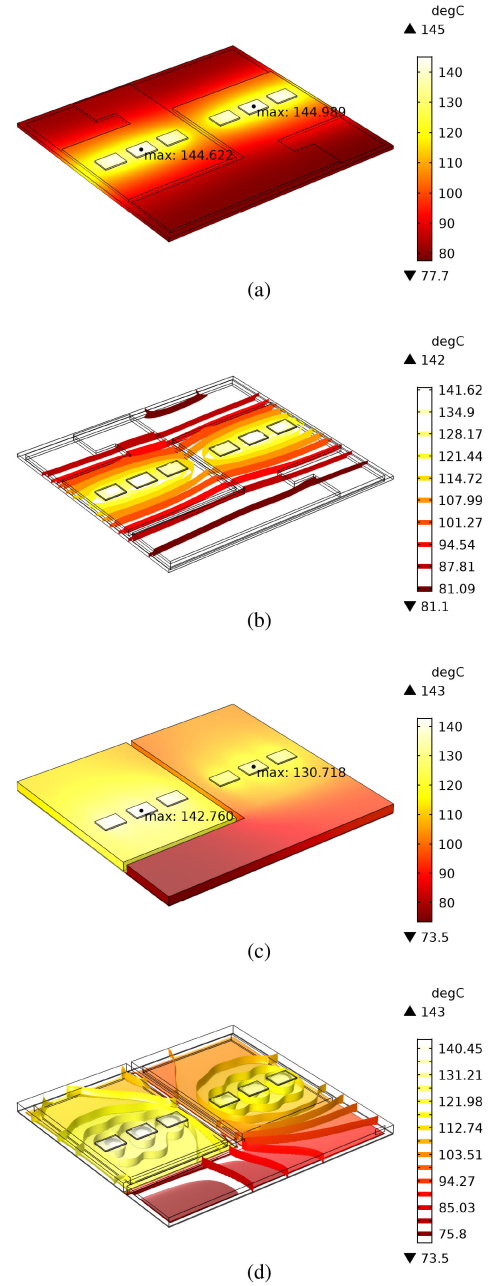


Fig. 9. Steady-state thermal analysis results for DBC and IMSwTPG. (a) Surface temperature distribution on DBC. (b) Isothermal contours in DBC. (c) Surface temperature distribution on IMSwTPG. (d) Isothermal contours in IMSwTPG.

where T_j is the recorded junction temperature at a given time instant, T_c is the base temperature, and P_{Loss} is the power loss value used to reach steady-state operation condition in the steady-state simulations. Transient thermal impedance curves for individual dies on DBC and IMSwTPG substrates are presented in Figs. 10 and 11, respectively. As shown in Fig. 10, the symmetrical thermal performance depicted in Fig. 9(a) was achieved via the well-matched steady-state thermal impedance of the SiC MOSFET dies on the DBC substrate. At steady-state conditions, the thermal impedance difference between six SiC MOSFET dies is less than 0.1 K/W. According to the transient

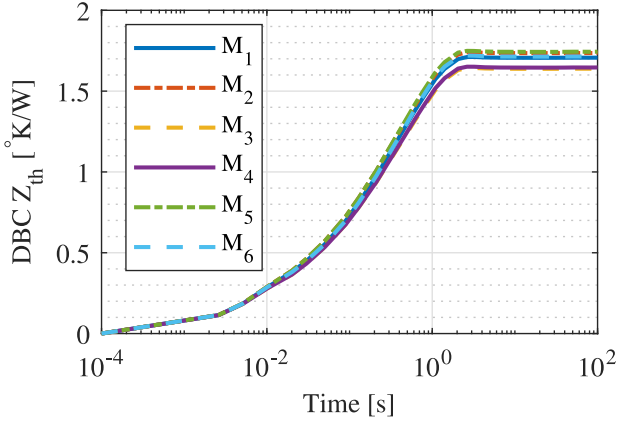


Fig. 10. Transient thermal impedance of each die on DBC.

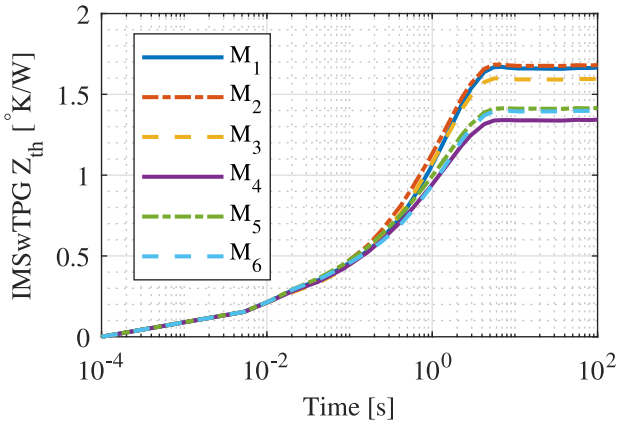


Fig. 11. Transient thermal impedance of each die on IMSwTPG.

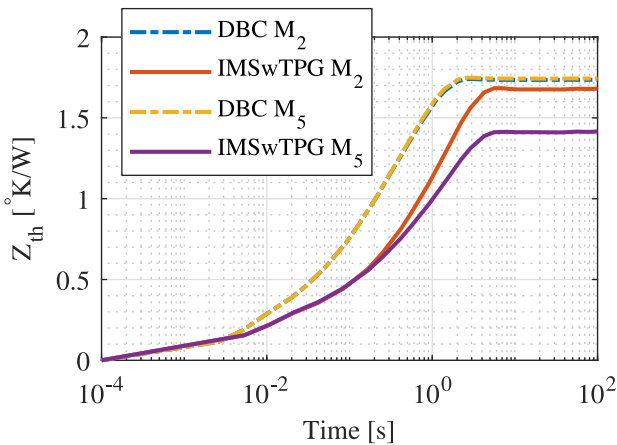
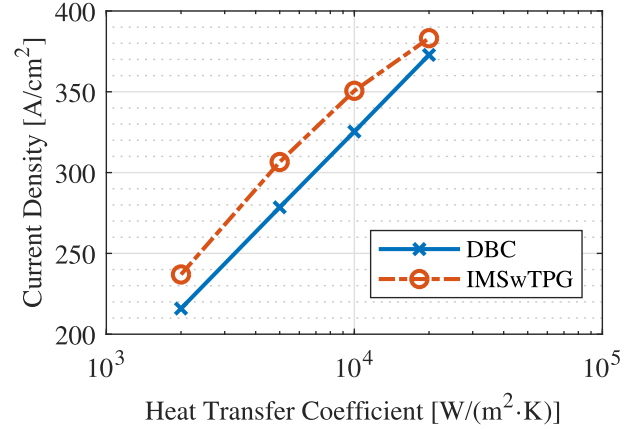


Fig. 12. Comparison of transient thermal impedance for M2 and M5 on DBC and IMSwTPG.

impedance results presented for the IMSwTPG in Fig. 11, the temperature difference at steady state between dies M_{1-3} and M_{4-6} can be explained with the steady-state impedance mismatch presented in Fig. 11. To compare the DBC substrate and the proposed IMSwTPG, the transient thermal impedance comparison of the most stressed dies M_2 and M_5 is presented in Fig. 12. The comparison results show that the IMSwTPG can

Fig. 13. Current density comparison for M_5 under different coolant performances.

outperform the DBC substrate at steady-state and transient conditions owing to enhanced heat spreading within the structure. For both M_2 and M_5 , the transient impedance between 10 ms and 1 s is improved more than 40%. For M_5 , the steady-state performance can be improved by 17%. The transient thermal impedance improvement will eventually lead to improved performance, especially for traction inverter systems in which the fundamental output frequency of the inverter has a direct impact on junction temperature variation [16]. The junction temperature variation can be reduced in the high-torque low-speed region by moving from a DBC to an IMSwTPG solution. Foster network parameters for both designs are extracted from simulation results in Figs. 10 and 11 and are presented in the Appendixes.

High-power modules can have a wide range of thermal performance based on the module structure, and also cooling strategy adopted. Conventional liquid-cooled cold plates can provide heat transfer coefficient varying from 1000 to 5000 W/(m²·K) depending on the internal design of the cold plate. In more advanced technologies such as direct jet impingement, as discussed in [6], [31], and [32], where the coolant is directly applied to the surface using jets, heat transfer coefficient values above 20 000 W/(m²·K) can be achieved. To quantify the impact of thermal impedance improvement for M_5 in terms of continuous current capability, the current density for the device M_5 , J_{M_5} , is calculated at different heat transfer coefficient conditions as follows:

$$J_{M_5} = \frac{1}{A_{\text{Die}}} \sqrt{\frac{T_j - T_c}{R_{\text{th}} \cdot R_{\text{DS(on)}}}} \quad (5)$$

where A_{Die} is the die surface area for CPM3-0900-0030 A ($4.08 \times 3.01 \text{ mm}^2$), T_j is the maximum junction temperature (150 °C), T_c is the base temperature (65 °C), R_{th} is steady-state thermal resistance, and $R_{\text{DS(on)}}$ is ON-state resistance at a junction temperature of 150 °C (39 mΩ according to the device data sheet). The current density results for M_5 with respect to different heat transfer coefficient values are presented in Fig. 13. It can be seen that, independent from cooling performance, the IMSwTPG can provide up to 10% increased current density by replacing the DBC substrate with an optimized IMSwTPG.

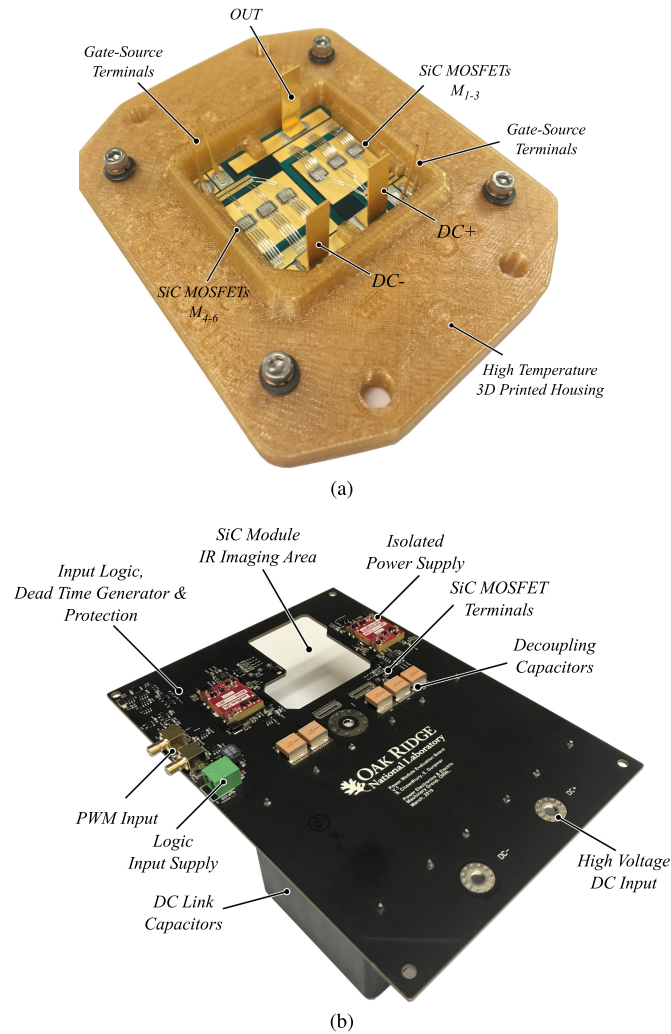


Fig. 14. (a) Half-bridge SiC MOSFET and IMSwTPG-based power module. (b) Power dissipation and loss analysis setup for thermal characterization of the substrates.

V. EXPERIMENTAL SETUP AND ANALYSIS

The experimental analysis was divided into two sections: 1) comparison of DBC and IMSwTPG based on steady-state and transient-based thermal characterization and 2) static and dynamic electrical characterization of a SiC-MOSFET-based IMSwTPG power module at rated operating conditions. Half-bridge SiC-MOSFET-based test samples were fabricated on DBC and IMSwTPG substrates, as shown in Fig. 14(a). The SiC MOSFETs were soldered onto the substrate with a 50- μm -thick 63-Sn/37-Pb solder, and the 250- μm Al wire bonds were used for power and gate interconnects. A high-temperature 3-D-printed structure was used for housing and assembly of the sample to the test setup.

For the thermal analysis, the substrate under evaluation was mounted on a liquid-cooled cold plate, with precalibrated thermocouples located at different locations on the base of the substrate, as shown in Fig. 15. High-performance thermal interface material with 13 W/(m $\cdot$ K) was used with 1-mm thickness between the substrate and the cold plate. The substrate surface

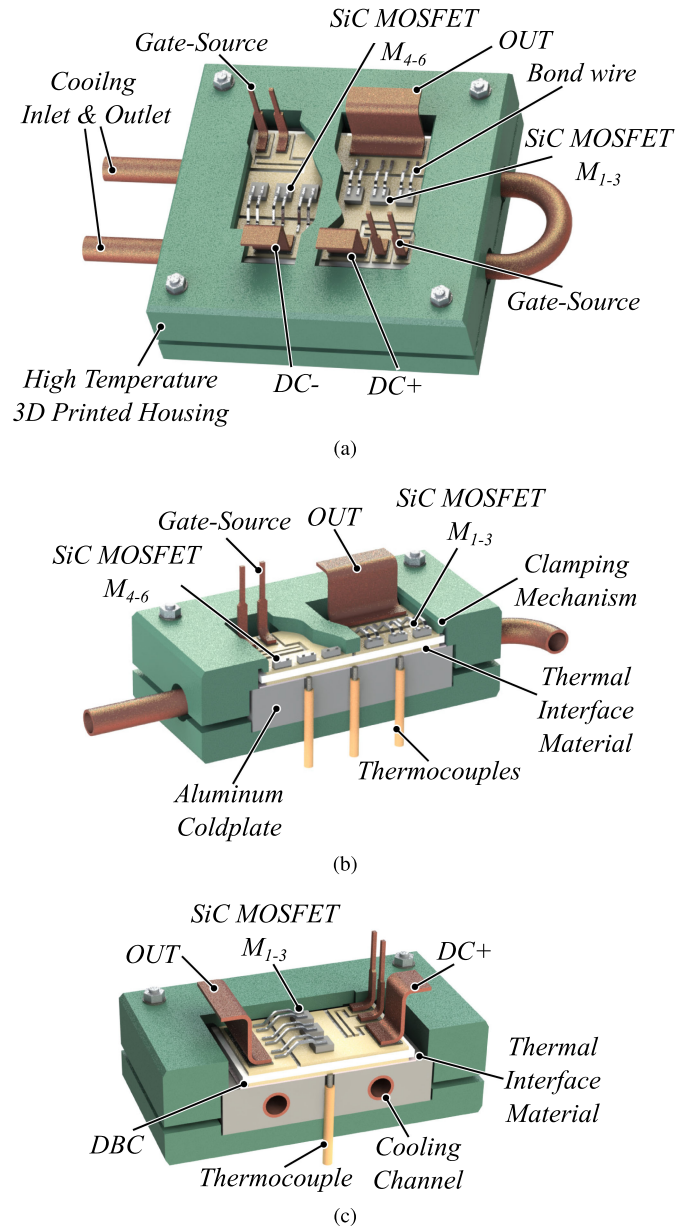


Fig. 15. Thermal measurement setup for DBC and IMSwTPG evaluation. (a) Isometric view. (b) Right cross section. (c) Front cross section.

and SiC MOSFET die surface temperatures were measured using an infrared (IR) thermal camera. For this purpose, the top surface of the sample was painted with black flat spray paint to achieve constant emissivity. The temperature measurement at the base of the substrate decoupled the thermal impedance measurement for DBC- and IMSwTPG-based samples, so the temperature difference between the SiC MOSFET die surface and the substrate base temperature could be directly measured.

The substrate sample was mounted on an evaluation board that provided isolated gate circuitry, protection, and high-power connection, as shown in Fig. 14(b). The experiments were conducted for both IMSwTPG and DBC substrates under the same operating conditions. The electrical setup shown in Fig. 16 was used for power dissipation and, therefore, thermal loading of

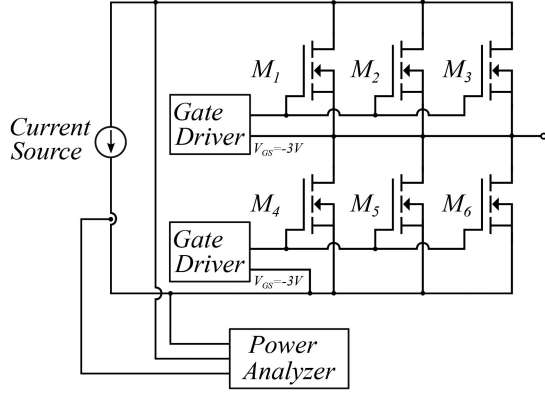


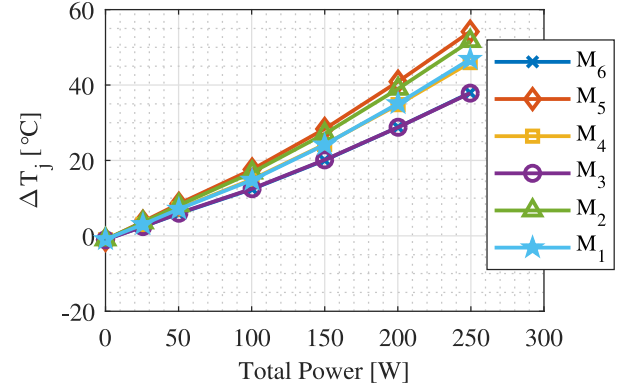
Fig. 16. Power dissipation and loss analysis setup for thermal characterization of the substrates.

the SiC devices on different substrates. During the experimental analysis, the SiC MOSFETs in the half-bridge configuration were turned OFF by applying -3 V to their gate. Then, a controllable dc source was connected to dc- and dc+ terminals on the substrate to dissipate power loss across the devices by utilizing relatively large ON-state voltage drop of body diode of SiC MOSFETs. The power losses across the devices M_{1-3} and M_{4-6} were measured by probing the current through the devices and the voltage across the substrate terminals via a power analyzer to ensure symmetrical power losses between top and bottom devices in the half-bridge. The total power dissipation across six SiC MOSFETs were swept from 0 to 250 W at three different coolant temperatures of 25, 45, and 65 °C conditions, and the flow rate and the coolant temperatures were regulated with a closed-loop chiller.

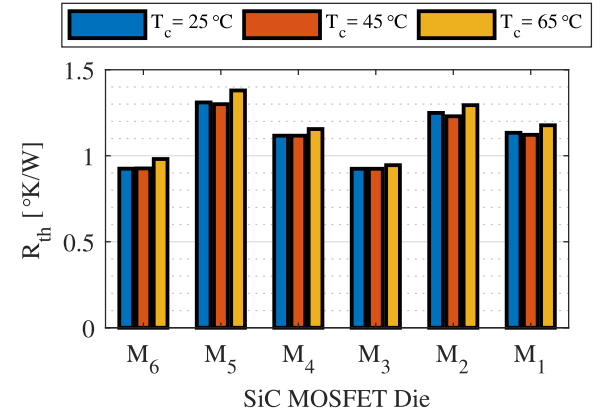
The electrical performance of the IMSwTPG was evaluated using conventional static (V_{DS} versus I_{DS}) and switching loss characterization based on the double-pulse setup [33].

A. Steady-State Thermal Analysis

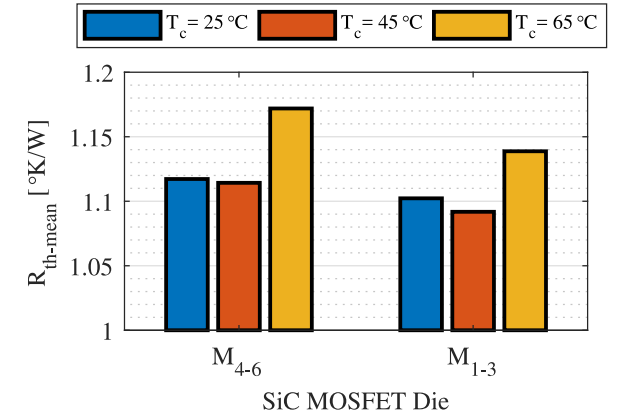
The steady-state thermal analysis was conducted by recording the average junction temperature T_j , the substrate base temperature T_c , and the power loss P_{LOSS} at a given operating point to calculate the thermal resistance based on (4). Junction temperature increase with respect to the total power loss across the six SiC MOSFET dies on the DBC substrate at 25 °C coolant temperature is presented in Fig. 17(a). The linear increase in ΔT_j with respect to power loss confirms that the melting temperature of the 63-Sn/37-Pb solder did not have an impact on thermal resistance measurements at high power loss. Based on the data in Fig. 10, steady-state thermal resistance for each die at three different coolant temperatures for the DBC-based sample is presented in Fig. 17(b). It can be seen that for each individual die, there was negligible variation of R_{th} with respect to coolant temperature, as expected. In contrast, there was a mismatch between paralleled dies M_{1-3} and M_{4-6} : this is related to the body diode conduction performance of the SiC MOSFET. For the heating of SiC MOSFETs, the body diode of the device is used to provide a large power loss across the device with reduced dc. This is achieved by utilizing the relatively large



(a)



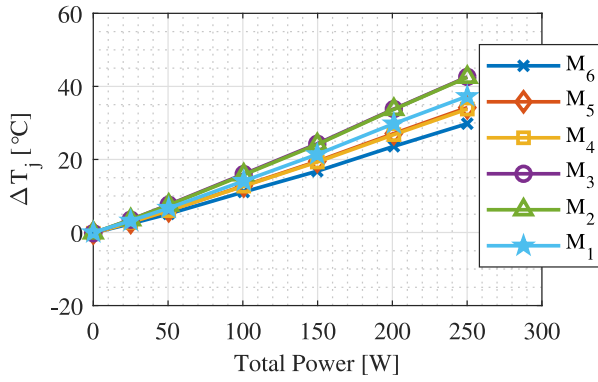
(b)



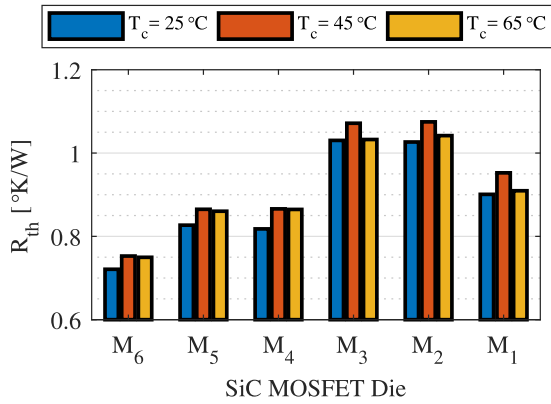
(c)

Fig. 17. (a) Junction temperature of individual SiC MOSFETs on the DBC substrate with respect to the substrate base at 25 °C coolant temperature. (b) Thermal resistance of individual SiC MOSFET with respect to the substrate base at different coolant temperatures. (c) Mean thermal resistance for paralleled SiC MOSFETs at different coolant temperatures.

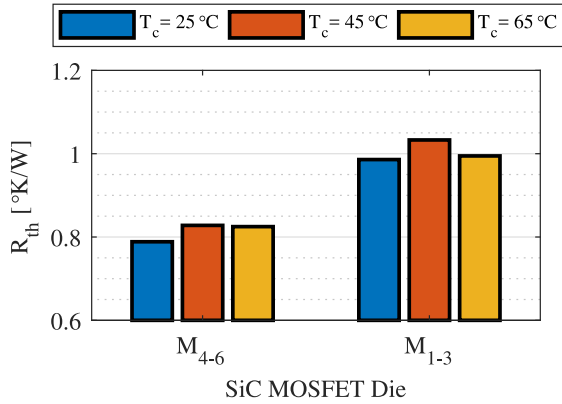
ON-state voltage drop of the body diode with respect to the device channel. However, according to the device data sheet provided by Wolfspeed [28], the SiC MOSFET body diode has a negative temperature coefficient, which is the probable cause for uneven power distribution across paralleled dies. As the experimental setup did not allow for measurement of individual die loss,



(a)



(b)



(c)

Fig. 18. (a) Junction temperature of individual SiC MOSFETs on the IMSwTPG with respect to the substrate base at 25 °C coolant temperature. (b) Thermal resistance of individual SiC MOSFET with respect to the substrate base at different coolant temperatures. (c) Mean thermal resistance for paralleled SiC MOSFETs at different coolant temperatures.

the average thermal resistance of paralleled dies $R_{th-mean}$ is presented in Fig. 17(c) at different coolant temperatures. The thermal symmetry of the DBC substrate is clearly seen by the comparison of $R_{th-mean}$ for M_{1-3} and M_{4-6} .

The IMSwTPG was evaluated for steady-state thermal performance under the same operating conditions as used in the DBC substrate analysis. The junction temperature increase of each SiC MOSFET die at 25 °C coolant temperature is presented in Fig. 18(a). Similar to DBC junction temperature results

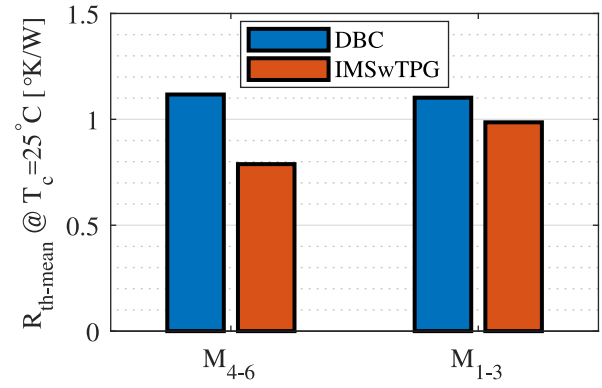


Fig. 19. Comparison of average thermal resistance of paralleled SiC MOSFETs with DBC and IMSwTPG substrates at 25 °C coolant temperature.

presented in Fig. 17(a), the ΔT_j measurements in Fig. 18(a) show linear increase with respect to power loss and validates the accuracy of thermal resistance measurements under high power loss. Steady-state thermal resistance of each die at three different coolant temperatures is presented in Fig. 18(b). As observed in the DBC sample, there is a slight asymmetry between paralleled dies M_{1-3} and M_{4-6} due to body diode performance. The average thermal resistance for paralleled dies in Fig. 18(c) presents the thermal resistance difference between M_{1-3} and M_{4-6} under three different coolant temperatures. The difference between M_{1-3} and M_{4-6} owing to the copper core difference, as observed in the FEA results shown in Figs. 9(c) and 11, is very clear. The temperature profile at 200-W total power loss and 25 °C coolant temperature for the IMSwTPG is presented in Fig. 20(b). The extended heat spreading for M_{4-6} provides 20% improvement in thermal resistance with respect to M_{1-3} , which is well aligned with the FEA results presented in Fig. 11.

The steady-state thermal performance comparison of DBC and IMSwTPG substrates is summarized with averaged thermal resistance of M_{1-3} and M_{4-6} at 25 °C coolant temperature in Fig. 19, and thermal images of DBC and IMSwTPG at 200-W total power loss are provided in Fig. 20(a) and (b), respectively. It should be noted that the temperature profiles presented for both cases are absolute temperature values measured with an IR camera. The clamping pressures applied to DBC and IMSwTPG samples are different and, therefore, have inherently different contact resistances to the aluminum cold plate shown in Fig. 15. The measurement of substrate temperature via thermocouples decouples the impact of the clamping effect to the comparison. Therefore, these images should be used in conjunction with the substrate base temperature to evaluate the structure's thermal resistance. Both images are scaled for a 100 °C range, and from the color distribution among the substrates, it can be concluded that lateral heat spreading worked effectively for the IMSwTPG structure.

B. Transient Thermal Analysis

Transient impedance extraction of the DBC and IMSwTPG substrates in the experimental setup is conducted via the junction temperature measurement of individual dies during the cool

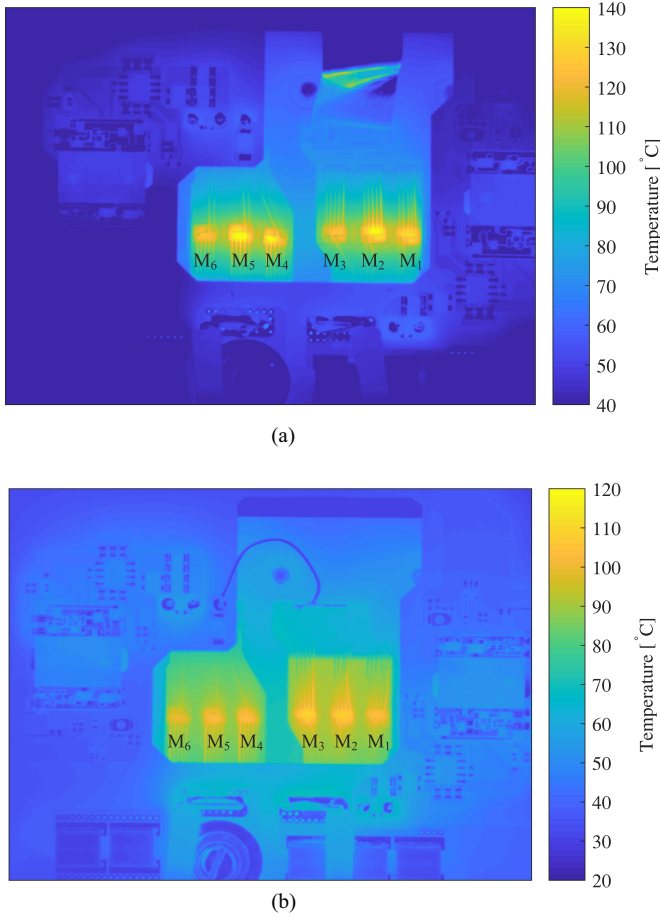


Fig. 20. Thermal images of (a) DBC substrate and (b) IMSwTPG at 200-W power loss and 25 °C coolant temperature.

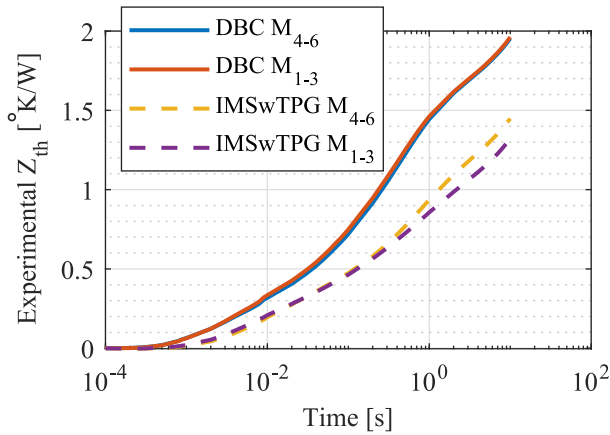


Fig. 21. Transient impedance of SiC MOSFET dies on DBC and IMSwTPG substrates.

down process at a 25 °C coolant temperature. Starting from 100 μ s, the junction temperatures of the dies were recorded with ten samples per decade sampling rate up to 10 s for extraction of thermal impedance [34]. The impedance results for paralleled dies M_{1-3} and M_{4-6} are presented in Fig. 21 for both DBC and IMSwTPG. In comparison to the FEA-based comparison shown

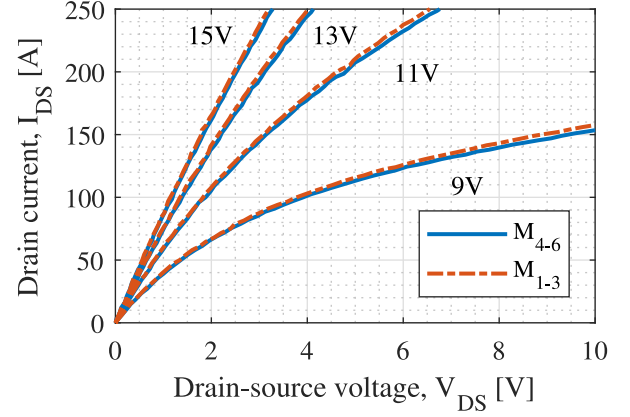


Fig. 22. Output characteristics of the switches M_{1-3} and M_{4-6} on the IMSwTPG at 25 °C junction temperature and different gate-source voltage conditions.

in Fig. 12, it can be concluded that the observed reduction in thermal impedance between 10 ms and 1 s is well matched with the FEA results. The impact of the reduction in thermal impedance is discussed for DBC and the conventional IMS by Gurpinar *et al.* [16]. At certain operating regions for electric drive traction inverters, in which maximum torque and low fundamental motor speed are desired, the reduced thermal impedance will provide lower junction temperature variation for SiC MOSFETs.

C. Electrical Characterization

The electrical characterization of the IMSwTPG-based power module was conducted at static and dynamic conditions for the top (M_{1-3}) and bottom (M_{4-6}) switches under different junction temperatures and load current conditions. In an experimental setup similar to that used for thermal characterization, the electrical characterization was performed using the evaluation board in Fig. 14(b) to provide the high-voltage dc input and the gate driver and auxiliary circuitry for successful operation of the power module shown in Fig. 14(a). The characterization was conducted at three different junction temperatures—25, 75, and 125 °C—to evaluate the conduction and switching performance of SiC MOSFETs on the IMSwTPG.

1) *Static Characterization:* The static characterization of the switches was conducted with a high-power curve tracer with up to 250-A drain current per switch. The SiC MOSFET dies used in this study were rated at 66 A at 25 °C junction temperature with 30-m Ω typical and 39-m Ω maximum ON-state resistance [28]. The output characteristics for M_{1-3} and M_{4-6} at a 25 °C junction temperature and different gate-source voltages are presented in Fig. 22. The output characteristics of both switches were matched well at different gate voltages. Based on the output characteristics extracted at different junction temperature and gate-source biases, the ON-state resistance for each switch was calculated as a function of drain current, as presented in Fig. 23. According to the data sheet of the die, the ON-state resistance of three paralleled dies should be between 10 and 13 m Ω at a 105-A drain current and a 25 °C junction temperature. At 125 °C, this ON-state resistance range should increase, by factor of approximately 1.2 m Ω to 12–15.6 m Ω . Considering only die conduction

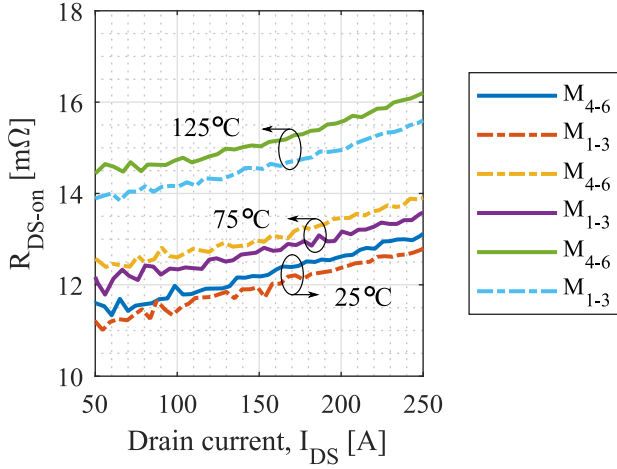


Fig. 23. ON-state resistance of M_{1-3} and M_{4-6} with 15-V gate-source voltage at three different junction temperatures.

performance, when these theoretical values are compared with the experimental results shown in Fig. 23—which include the contribution of wire bond, terminal, and substrate performance to ON-state resistance—it is clear that the IMSwTPG—including the TPG tiles, the Cu core underneath the dies, and the interconnects used in the module design—had minimal impact on conduction performance. The measured ON-state resistance presented in Fig. 23 is very close to the theoretical value.

2) *Dynamic Characterization*: The dynamic characterization was conducted in a conventional double-pulse setup [35]. The load inductor was connected across the top switch M_{1-3} and was used as the freewheeling element when the bottom switch M_{4-6} was OFF. The M_{4-6} was hard switched during turn-ON and turn-OFF, and the drain-source voltage V_{DS} and drain-source current I_{DS} were recorded at different load current and junction temperature conditions. Test parameters for the double-pulse characterization are presented in Table V in the Appendixes. The drain-source voltage and drain current waveforms at 125 °C junction temperature, 400-V dc link, and 90-A load current conditions for turn-ON and turn-OFF transients are presented in Fig. 24(a) and (b), respectively. The turn-ON and turn-OFF transition times were less than 50 ns, which validates the high performance of the proposed architecture. Turn-ON (E_{on}) and turn-OFF (E_{off}) switching energies under different load currents and three different junction temperatures are presented in Fig. 25. The turn-ON and turn-OFF loss energies are independent of junction temperatures, which were previously reported for TO-220 packaged 650-V SiC MOSFETs by Gurpinar *et al.* [36]. Furthermore, the turn-OFF energy increased gradually beyond 100-A load current, and one cause of the increased turn-OFF loss was the voltage overshoot caused by loop stray inductance, which was estimated at around 15 nH based on the voltage overshoot and current slope in Fig. 24(b). The wire bonds are known for large stray inductance, and several wire bondless power module solutions for WBG devices have been discussed in literature [4]. Although the electrical layout used in this design was to match the DBC layout for fair comparison between the two, the multilayer structure of the IMSwTPG allows for

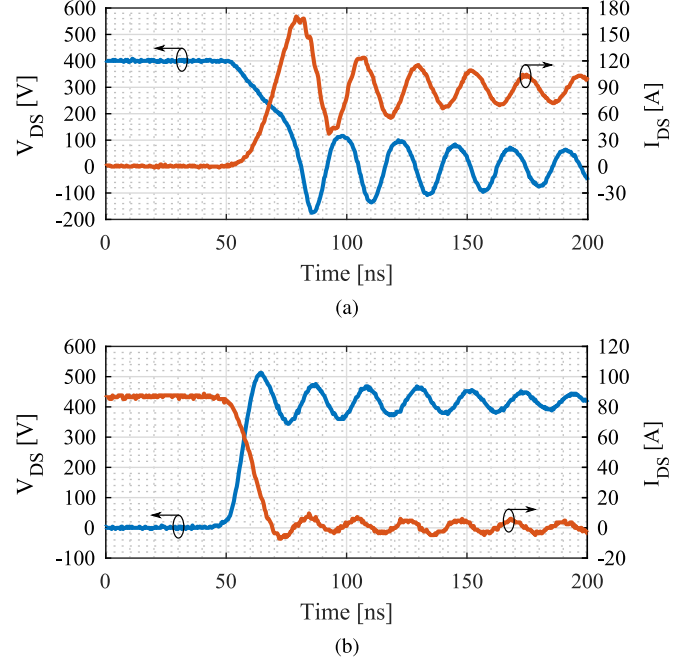


Fig. 24. (a) Turn-ON and (b) turn-OFF switching waveforms at 400-V dc link, 90-A load current, 125 °C junction temperature.

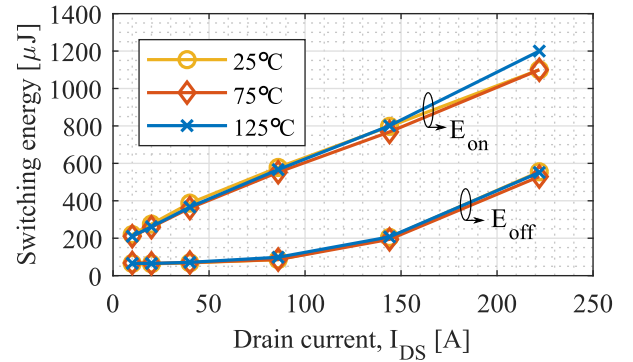


Fig. 25. Turn-ON (E_{on}) and turn-OFF (E_{off}) energy of M_{4-6} with respect to the drain current at 400-V dc-link voltage.

creating optimized layouts for WBG-based power modules to minimize stray inductance in the commutation loop.

VI. CONCLUSION

This article presents a graphite-embedded IMS designed for WBG power modules. The graphite-embedded Cu core, which is the fundamental building block of the graphite embedded substrate, outperforms bulk Cu because of very high thermal conductivity of graphite. The anisotropic properties of graphite can be used to enhance the lateral heat spreading across the power module and the heat transfer from the die to the coolant. It is shown that by using graphite-embedded substrates, the junction-to-case thermal resistance of SiC MOSFETs can be reduced up to 17%, and the device current density can be increased by 10%, regardless of the thermal management strategy used to cool the substrate. Furthermore, the transient thermal impedance reduction of up to 40% of the dies owing to increased heat

TABLE III
FOSTER NETWORK THERMAL PARAMETERS FOR DIES ON DBC

Impedance i		1	2	3	4	5
M_1	R_{thi} [$^{\circ}\text{K/W}$]	1.037	0.0001043	0.2847	0.3868	0.0008247
	τ_i [s]	0.5348	0.1801	0.006643	0.08564	0.7555
M_2	R_{thi} [$^{\circ}\text{K/W}$]	0.4384	0.5196	0.2769	0.4957	0.007333
	τ_i [s]	0.08641	0.5562	0.006492	0.5289	0.02078
M_3	R_{thi} [$^{\circ}\text{K/W}$]	0.003954	0.3663	1.01	0.0003221	0.2601
	τ_i [s]	0.1235	0.07553	0.5345	0.6592	0.00595
M_4	R_{thi} [$^{\circ}\text{K/W}$]	0.2427	0.9493	0.3726	0.08221	0.001436
	τ_i [s]	0.005379	0.5256	0.06857	0.5236	0.541
M_5	R_{thi} [$^{\circ}\text{K/W}$]	0.3019	0.0001493	1.018	0.1443	0.2818
	τ_i [s]	0.007285	0.5903	0.5376	0.114	0.08627
M_6	R_{thi} [$^{\circ}\text{K/W}$]	0.1034	0.1858	0.07542	0.2737	1.078
	τ_i [s]	0.003498	0.009938	0.0836	0.07754	0.5159

TABLE IV
FOSTER NETWORK THERMAL PARAMETERS FOR DIES ON IMSWTPG

Impedance i		1	2	3	4	5
M_1	R_{thi} [$^{\circ}\text{K/W}$]	0.1998	0.3199	0.4343	0.3827	0.3298
	τ_i [s]	0.507	1.496	1.382	1.371	0.01088
M_2	R_{thi} [$^{\circ}\text{K/W}$]	0.2458	0.002143	0.02156	0.286	1.126
	τ_i [s]	0.2048	0.9108	1.044	0.008647	1.328
M_3	R_{thi} [$^{\circ}\text{K/W}$]	0.09143	0.2494	0.1976	1.059	0.0007677
	τ_i [s]	0.1045	0.00683	0.3231	1.343	1.054
M_4	R_{thi} [$^{\circ}\text{K/W}$]	0.728	0.1145	0.001145	0.24	0.2583
	τ_i [s]	1.337	1.366	0.3292	0.1419	0.007329
M_5	R_{thi} [$^{\circ}\text{K/W}$]	0.2534	0.1414	0.2818	0.05887	0.6796
	τ_i [s]	0.007076	1.143	0.13	1.384	1.403
M_6	R_{thi} [$^{\circ}\text{K/W}$]	0.001577	0.2606	0.1853	0.0008965	0.9509
	τ_i [s]	0.003895	0.00771	0.09065	1.352	1.373

TABLE V
DOUBLE-PULSE TEST PARAMETERS FOR DYNAMIC CHARACTERIZATION

Parameter	Symbol	Value
DC Link Voltage	V_{DC}	400V
Load Inductance	L_{Load}	40 μH
Junction Temperature	T_j	25 $^{\circ}\text{C}$, 75 $^{\circ}\text{C}$, 125 $^{\circ}\text{C}$
External turn-on gate resistance	R_{on}	3 Ω
External turn-off gate resistance	R_{off}	1 Ω
Drain Current	I_{DS}	10A to 225A

capacity is validated in transient thermal simulations and experiments. The electrical performance of the IMSwTPG-based SiC power module is validated by the static and dynamic characterization under high current and voltage conditions. The graphite core has virtually no impact on the ON-state resistance and switching performance. The overall simulation and experimental analysis of the IMSwTPG show that the IMSwTPG is an attractive solution for WBG-based power modules in terms of design flexibility, as well as static and transient electrical and thermal performance. As part of future work, optimization of IMSwTPGs will be studied for achieving high power density and thermal performance for WBG-based power modules.

APPENDIX I

FOSTER NETWORK PARAMETERS AND DOUBLE-PULSE TEST PARAMETERS

The Foster network thermal parameters for SiC MOSFET dies on DBC and IMSwTPG are presented in Tables III and IV, respectively. The parameters can be used in the following equation

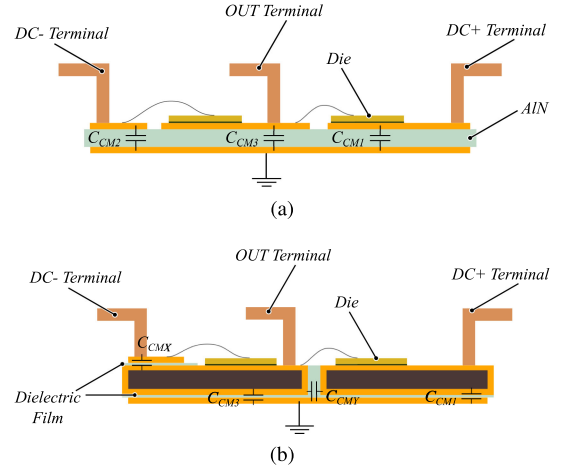


Fig. 26. Parasitic capacitances in (a) DBC and (b) IMSwTPG substrates.

TABLE VI
PARASITIC CAPACITANCE VALUES FOR DBC AND IMSWTPG

DBC		IMS	
C_{CM1}	64pF	C_{CM1}	282.4pF
C_{CM2}	27pF	C_{CM3}	418.16pF
C_{CM3}	92pF	C_{CMx}	2.74pF
		C_{CMY}	77.4pF

for Foster-network-based thermal models for each die:

$$Z_{th} = \sum_{i=1}^5 R_{thi} \cdot (1 - e^{-t/\tau_i}). \quad (6)$$

APPENDIX II

PARASITIC CAPACITANCE COMPARISON OF DBC AND IMSWTPG

The IMSs usually suffer from the parasitic capacitance due to reduced thickness of the dielectric, as highlighted in [37]. The parasitic capacitance for DBC and IMSwTPG are presented in Fig. 26(a) and (b), respectively. The parasitic capacitors between the live terminals and the ground are charged or discharged based on the common-mode voltage observed across the capacitor terminals. In half-bridge circuits, as the output voltage with respect to ground is alternating between $+V_{DC}/2$ and $-V_{DC}/2$, significant current can flow through C_{CM3} depending on the dc-link voltage, the voltage slew rate, and the capacitance value. Both DBC and IMSwTPG structures were analyzed using the ANSYS Q3D FEA tool for parasitic extraction, and the parasitic capacitance values at 1 MHz for the components shown in Fig. 26 are presented in Table VI.

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